Automated Layman Abstractive Summarizer (ALAS) of C.S Research Papers

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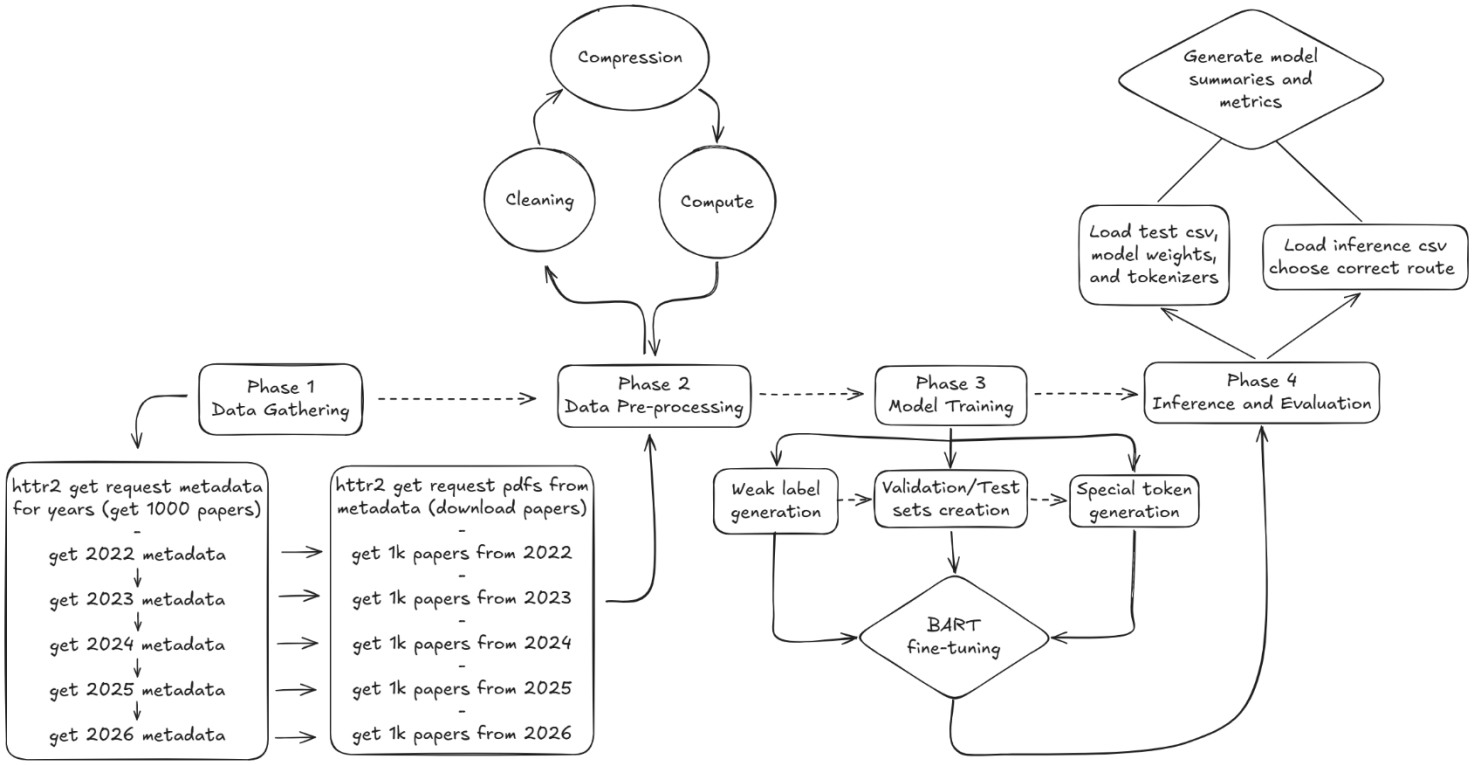
Introduction

The general public lacks active engagement in academic literature, while blame can be discussed as macro mechanisms, and assigned to algorithms that prey on individual's brain chemistry such as social media, which has been linked to negative effects in peoples' lives[1]; we instead propose a solution based on engagement metrics. For starters, shortening and simplifying text in news has been proven to promote reader engagement with the material[2], this is the core idea of our proposal and should theoretically alone be sufficient for promoting reading of advanced research papers. Furthermore, this does not just apply to news articles and magazines, similar studies were performed to check the validity of these approaches in more scientific areas, and the accuracy of the claims were confirmed[3]. Lastly, a side-effect of this phenomenon of read easiness was increased agreeability[4], while this may appear helpful short-term in a period of increased anti-scientific community popularity[5], we must be cautious of the long-term effects this may cause in people's subservience to authority. Nevertheless, we explored a solution involving deep learning, artificial-intelligence (AI) for creative text generation while relying on lightweight natural language processing (NLP) algorithms for tasks such as preprocessing, and alignment; thus, reducing computational demands, providing feasible solutions to the general public who may not have access to state-of-the-art hardware for AI optimization, as well as minimizing environmental damage. For this purpose, a nuanced approach in system

design and architecture had to be applied to reduce AI tasking as much as possible. An overview of the workflow of the code in its totality is present in Figure 1 below.



First, we gathered 1000 papers from the years 2022 to 2026 (5k total papers) along with useful metadata such as main author, and abstract. Then, we pre-processed the data in accordance with the results found in “The extraction of a brief summary from scientific documents using machine learning methods".[6] which found that abstractive summarizers often performed better when trained on domain-specific knowledge; additionally, this paper along with others have found the data-cleaning step to be crucial for better results, thus we will also be implementing this at this point. Once the data was ready, we performed fine-tuning (model training) of a larger model. Lastly, we compared our model and architecture with others to validate the effectiveness of our approach.

Methods

Phase 1

The pipeline was divided into four main phases. The first phase of data gathering involved querying requests to the arXiv database[7] to gather metadata of computer science (CS) specific research papers from any field within CS, we then processed the atom-formatted

XML metadata responses and began downloading the thousand papers for that specific year while making sure to abide by arXiv's request-time guidelines as to prevent rate-limiting. Lastly, we combined the metadata files (csv) into a singular tabular dataset to avoid recursively gathering each during pre-processing (this would increase computation time of NLP algorithms). Additionally, before moving on to phase 2, we gathered main-author demographic data from Namsor who was responsible for providing enriched csvs containing the paper author's gender, diaspora (ethnicity), and country of residence based on AI confidence scores. As Namsor is a paid service, we were limited by free accounts to only gather this data from 10% of authors (500/5000).

Phase 2

This phase involved the heavy lifting of data processing so the data looked as clean as possible while being appropriately formatted for use in model training down the road. To achieve this, heavy use of regular expression (regex) ruling and some lru caching was implemented as to reduce computation as much as possible. First, a starting step was applied to perform extensive pre-formatting such as Unicode normalization, citation and reference stripping, detection and removal of lines that looked like page numbers, URLs, bibliography text, etc. The goal here is not to perfectly reconstruct a perfectly clean version of the original, but rather produce something that can easily be fed into TextRank for compression. We also included functions to help deal with glued tokens though not perfectly, these appeared during pdf extraction due to relying on the straight forward pypdf package which warns about pdf extraction being unreliable[8]; indicating that data processing could be refined to either rely on machine learning algorithms designed for pdf extraction and parsing such as Generation of Bibliographic Data (GROBID), or more algorithmic work needs to be spent finding good rules and regexes for separating glued tokens.

Secondly after data cleaning, the following step was section extraction, for this purpose the code attempted to detect paper headings such as abstract, introduction, methods, etc. based on a combination of regex matching, alias lists, and heuristic checks for section-like titles. Upon section identification, the script generated section digests (mini-summaries), and a global digest (made from a mixture of recoverable sections plus abstract, and in some

worst-case scenarios, truncated raw cleaned pdf text) using Biased TextRank[9], an unsupervised, graph-based content extraction algorithm that excels at focused summarization by prioritizing nodes that are more relevant to our specific bias i.e. section terms. This sectioning design is purposeful as later stages will rely on a Map-Reduce approach for generating paper summaries, the skeleton for that feature is created here. Furthermore, we also computed important values like the route a summary should go, along with confidence on that route, coverage, compression ratio, etc. not only are these values crucial for inference, but we also want to quantify the "quality" of our data after pre-processing as to prevent introducing noisy data into model training; to this extent, data is assigned "good" or "borderline" if it can be used for training, and "bad" if it is too noisy and excluded from training; after quality assignment, our final total trainable dataset consisted of 4600~ papers and ~400 removed papers, this means we lost about ~8% of data during pre-processing which is acceptable and an improvement from earlier versions which resulted in a loss of ~25% of data.

Lastly, another important feature of this phase was computing readability metrics such as Flesch Kincaid grades, Zipf frequency, word count, among others. Diagnosing readability was not only important due to the project goals such as model comparisons, but it also proved effective in discovering outliers which could be considered again during cleaning or removal.

Phase 3

The training stage was the crux of the architecture; creative and effective system design would be the difference between the model bottlenecking or producing desirable results. A limiting factor in our data beyond any code-based noise introduction was our validation and test dataset; we manually prompted LLMs for a set of summaries from 50 random papers in our total dataset, to be used for validation (10), and testing (40). This summary set is referred to as the "silver-set"; as opposed to golden standard summaries which would be human authored and human reviewed, this silver set was purely LLM generated and only human reviewed. Due to time constraints and field limitations, we were neither able to gather already created human-summaries unlike many medical research papers which supply such summaries, nor had the time to read 50 research papers and create summaries.

Since we did not rely on manually labeled large-scale lay summary datasets, we applied a weakly supervised multitask formulation. Firstly, we assign merge keys for both pre-processed and silver-set data so that validation-testing papers are removed before synthetic training examples are generated, and then added back based on the keys. Secondly, we created pseudo-targets for section tasks using a creative approach of presenting paper abstracts as basically the author creating their own summary for the entire paper. Following this assumption, we aligned certain sentences in the abstract to a specific section using Term-Frequency Inverse Document-Frequency (TF-IDF)[10] vectorization (with added position and keyword bonuses), and cosine similarity. This approach intentionally kept the weak labels imperfect but grounded, the goal was to produce some form of supervision to nudge the model toward layman-like phrasing whilst preventing hallucination and straying too far from paper content/structure.

Then, the training set is built as a multitask collection of prompt-target pairs. The prompts define a special task based on sections such as [INTRO], [FINAL], [GLOBAL], etc. This task takes a section digest as input and maps it to a pseudo lay summary of that section while global maps a global digest to a pseudo full-paper summary, and final maps a stitched version of the mini-summaries into a full final summary, recreating the Map-Reduce format. This architecture is crucial for matching the intended inference behavior, the model is not asked to just write one kind of summary, instead it is trained to perform the same family of subproblems and learn to tackle the best route. After the training data has been built, we perform the actual fine-tuning using Performance-Efficient Fine-Tuning (PEFT) with Low-Rank Adaptation (LoRA)[11] on Facebook's BART model fine-tuned on CNN Daily Mail. BART is a large transformer sequence-to-sequence model with a bidirectional (BERT-like) encoder and an autoregressive (GPT-like) decoder, and is particularly effective when fine-tuned for text generation such as summarization[12]. The special section tokens were added to the tokenizer, token embeddings were resized to fit these special tags, and then the model was wrapped with LoRA, targeting the query state and value state modules; q_proj to decide what to pay attention to, and v_proj to decide what to carry forward. Then, Recall-Oriented Understudy for Gisting Evaluation (ROUGE) was used during validation to

evaluate N-gram overlap as a test of accuracy, and both the final checkpoint and tokenizer artifacts were saved.

Phase 4

In the last phase of our project, we perform simple inference of the fine-tuned model (run the final trained model on the best checkpoints/weights). Then, after picking the best checkpoint for our model, we were ready to run it against the other off-the-shelf (base) models. First, our model looks at route hints and section route confidence to make a choice of how to behave during summarization. We apply a Map-Reduce style approach where the model will create specific summaries for each mini-section, this is the Map step; then it will gather all the mini-summaries and it creates a final (that's where [FINAL] comes from) summary, this is the Reduce step. If the route confidence is low or the paper is missing too many section digests, the model will fall back to the [GLOBAL] tag and just use a mix of the abstract plus whatever is usable from the sections or paper.

Then, we load the same test csv for a fair comparison of all models, load the weights, load tokenizers and we fit our special tokens like [INTRO], [METHODS], [FINAL], etc. into the other models and then resize their token embedding matrix to turn our labels into legal inputs for the other models. After everything is set up and ready, we run inference on the models on the test csv set and record/compute model scores and readability metrics for model comparison.

Results

During metadata analysis, we performed demographic analysis of 10% of main authors based on the previous features described as gender, ethnicity, and country of residence. Figures 2, 3, and 4 below presents our findings for these features.

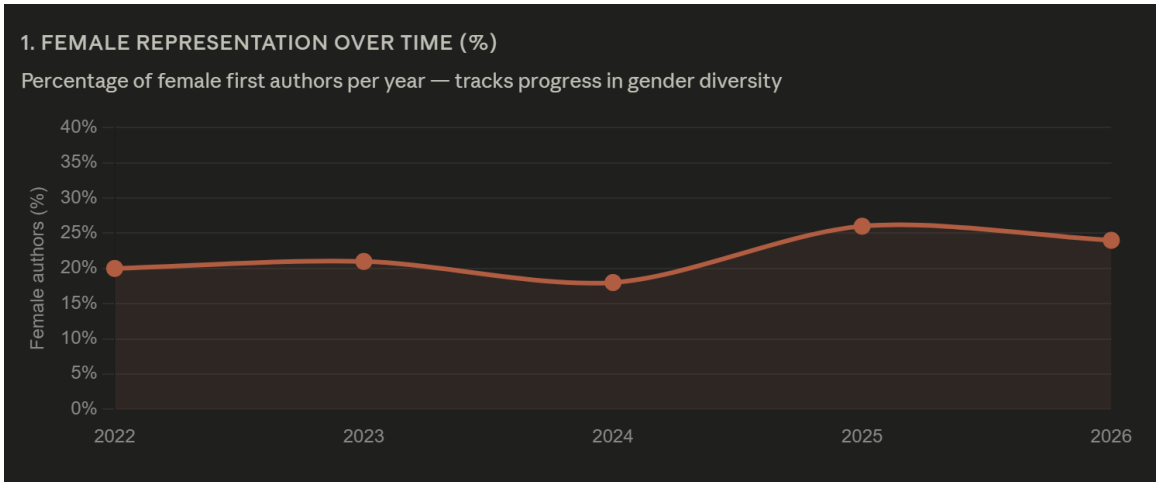
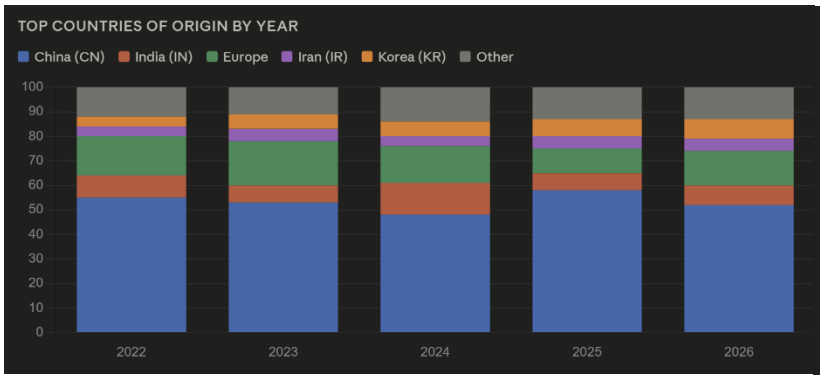
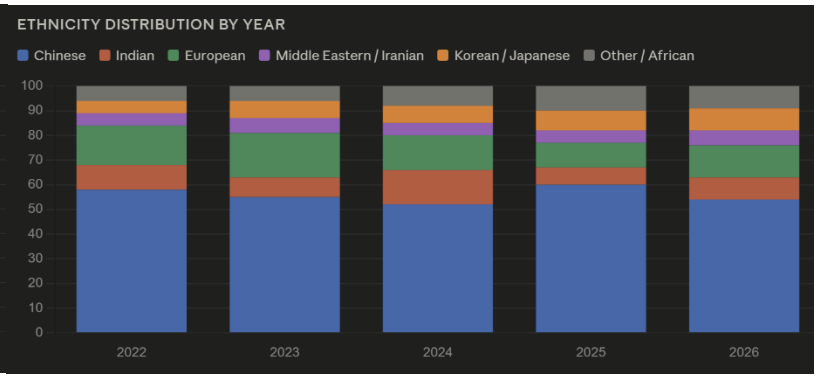


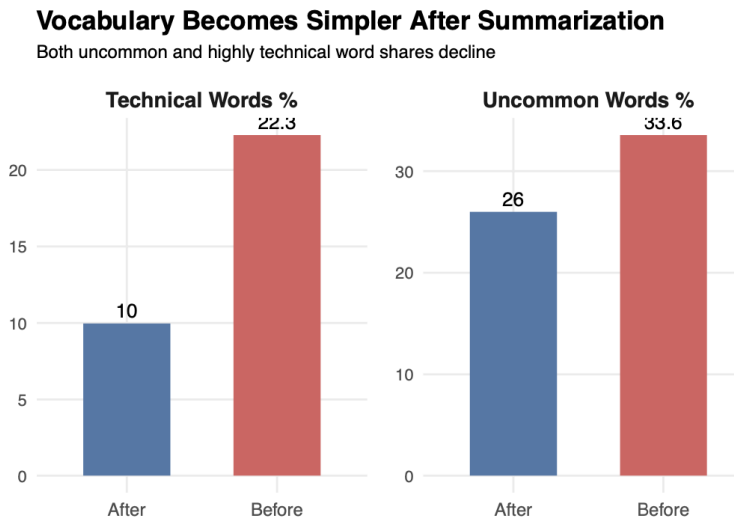
Figure 2 suggests female authorship increase overtime from 20% on 2022 to 25% on 2026.

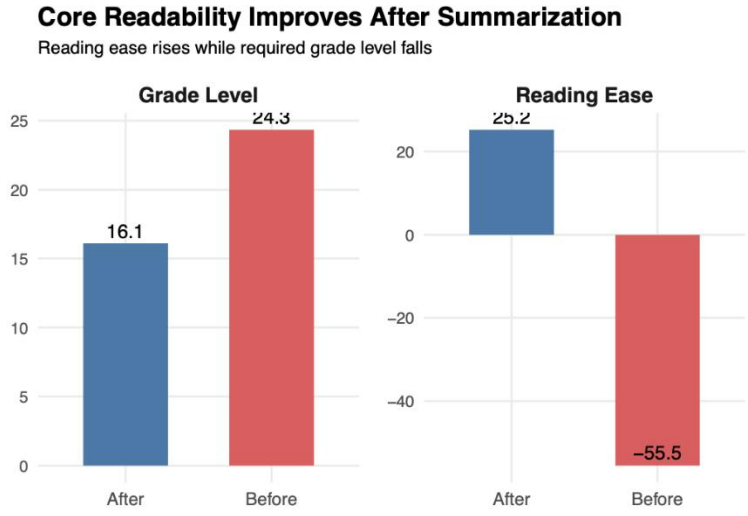




Figures 3 and 4 indicate China dominating in research paper production. However, the most interesting statistic and reason for extraction of these two distinct features of ethnicity and country of origin is due to the difference in values. The extraction tool and data pool is not perfect, but this does provide insight into the need of proper analysis of these features to better understand the difference of where papers are coming from and where their authors are coming from which are two different questions.

Next, during phase 2 we measured the difference in our dataset's readability before and after compression, to get a better understanding of compression techniques and how to properly apply them. In the end, we achieved great results presented by Figures 5 and 6 below.





Figures 5 and 6 above discover the change in readability during compression. Properly applying Biased TextRank with section focuses promote reading ease while reducing Flesch Kincaid grade level, this means that if we consider a high-school senior as grade 12, the compressed research papers go from requiring PhD-level, field-expertise reading, down to undergrad. In addition, the number of technical words is reduced by half while also promoting less uncommon word frequency.

Finally, Figures 7, 8, 9, 10, and 11 project the model's evaluation compared to other large base models such as Facebook's (base) BART, Google's Pegasus-arXiv, and AllenAI's LED-base-16384.

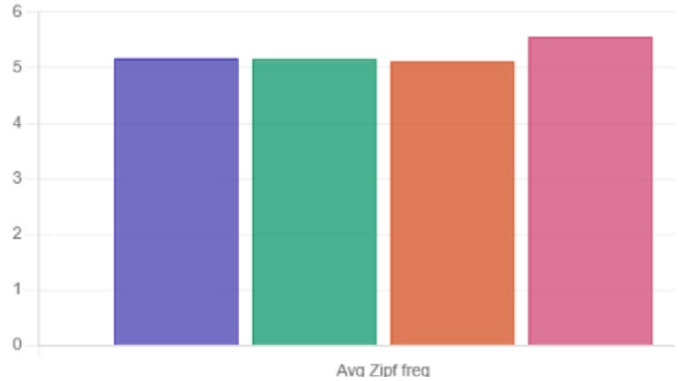


Figure 7 shows the average zipf frequency or average word commonness, our model is tied with base BART and LED at having the most common/everyday vocabulary.

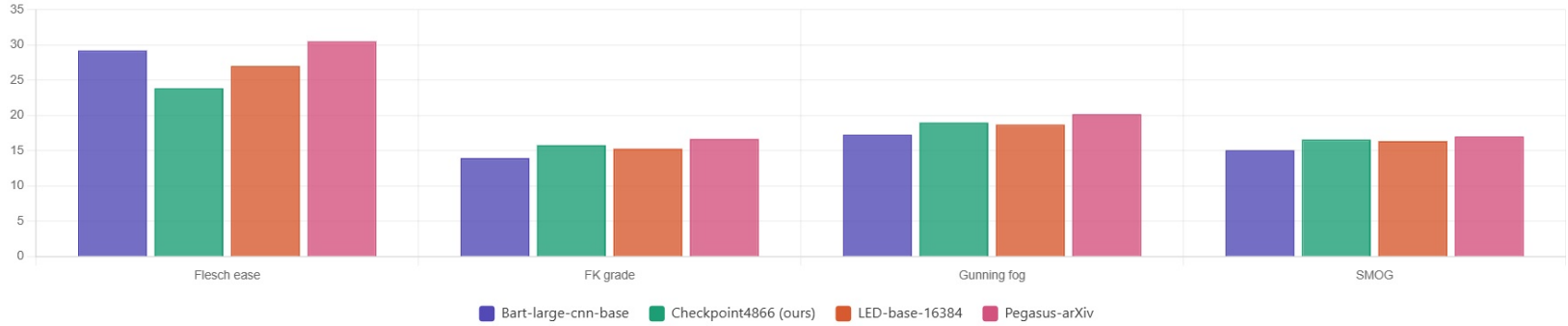
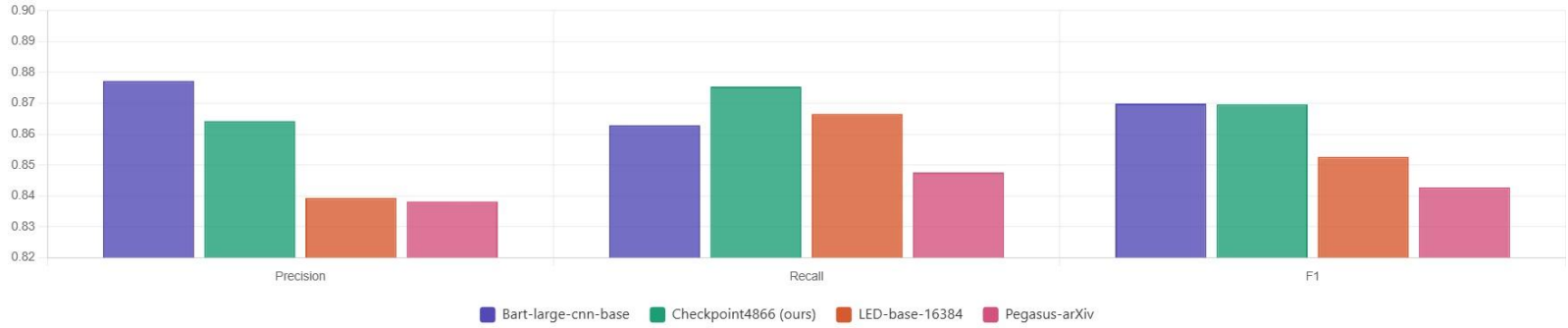
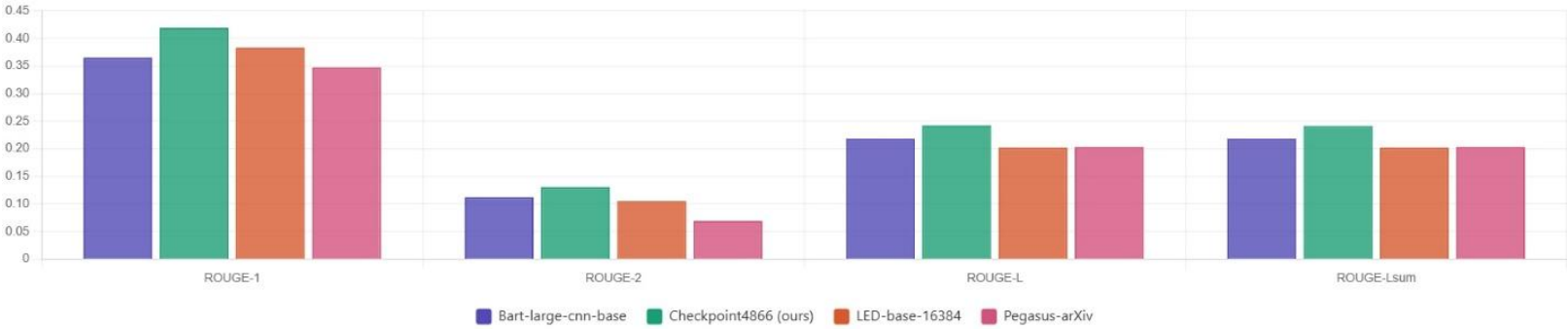


Figure 8 above shows our model being competitive at measurements like gunning fog, smog, and Flesch Kincaid which are all grade level type difficulty indexes. However, our model appears to fall short at simple Flesch ease while being easy to read on other metrics; this indicates that the really negative outliers in Flesch ease are bringing our model average grade down heavily in this specific metric.

Figure 9 below shows the BERT scores between the models which indicate how close the model summaries stay to the overall meaning (not hallucinating or confusing terms) of the

original paper i.e. semantic similarity. Additionally, Figure 10 shows the ROUGE scores between the models which indicate how close the model summaries stay to the overall structure (not removing or mixing up sections) of the original paper i.e. lexical similarity.





Figures 9 and 10 above indicate our model is tied with the base BART model at accuracy (BERT-F1), as well as beating all other models at N-Gram word overlap (ROUGE-Lsum).

Finally, Figure 11 below shows the compression metrics of our model compared to the rest.

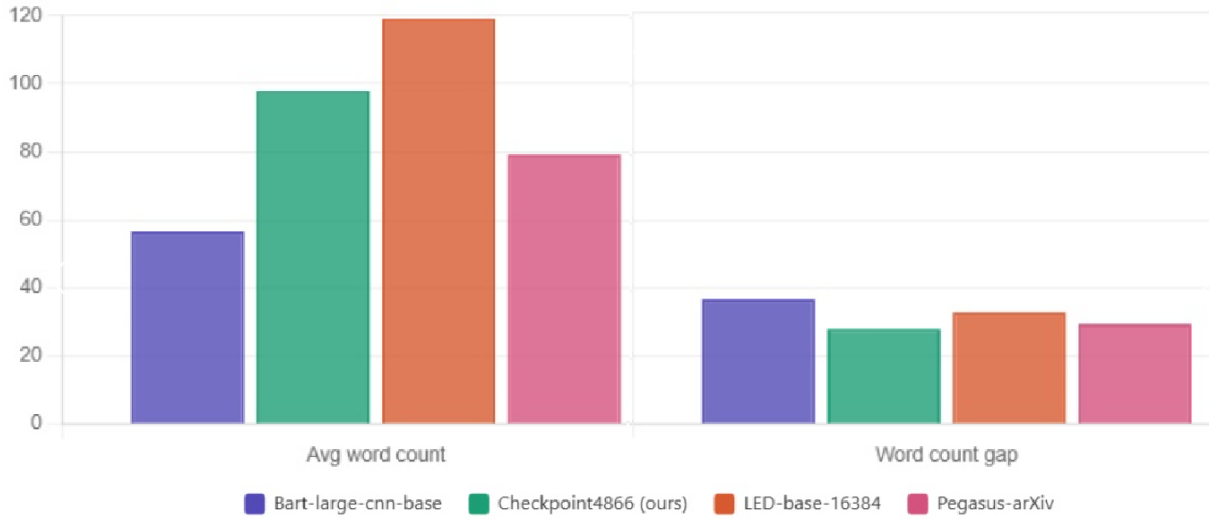


Figure 11 above suggest improvement, though the better suited Pegasus and base BART beat our model at this task.

Conclusion

In conclusion, this paper proposes a methodology for designing system architectures for abstractive research paper summarizers, rather than relying on single large models; the project built a complete pipeline that handles inputs, de-noising, and task-specific outputs from start to end. This design made it possible to explore a solution that is both accessible and achievable by being for the layman, and by the layman. At the same time, the project revealed the system's limitations. Namely, section summary repetition proving flawed section label overlap, silver evaluation sets not being gold standard human-adapted strong-supervised lay summaries, and readability metric comparison with models not accounting for heavy outliers created during imperfect compression, as well as obviously comparing models designed and optimized for different tasks rather than single handedly training every model being compared on the same architecture/design. However, by no means do these limitations invalidate our system, as the structured pipeline has proven to stay competitive and provide a great baseline in bridging the gap between technical research papers and accessible summaries for non-expert readers.

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